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ABSTRACT

Stuttering is a speech disorder that disrupts the natural flow of communication through involuntary repetitions, prolongations, blocks, and pauses in speech. Nearly 1% of people worldwide are impacted, and it can significantly affect social, professional, and personal life. Traditional diagnosis methods rely on manual assessment by speech-language pathologists, which are subjective, time-consuming, and not easily accessible to all individuals. To overcome these limitations, this project presents an intelligent and automated system that leverages deep learning to detect and classify various types of stuttering events in real-time speech recordings.

The core of the system is a multi-model Convolutional Neural Network (CNN)-based architecture, where five parallel binary classifiers are individually trained to detect distinct stutter types — including sound repetitions, word repetitions, prolongations, blocks, and interjections. The input audio is preprocessed using techniques such as segmentation, and Mel-spectrogram extraction to capture the essential temporal and spectral features of speech. Each CNN model processes these features independently and outputs probability-based predictions for specific dysfluencies. The system is integrated with a user-friendly PyQt5 graphical interface that allows users to record or upload audio samples, visualize waveform and spectrograms, and receive detailed analysis and classification reports in real time.

By combining deep learning-based speech analysis with an interactive application interface, this system provides a reliable and accessible tool for early stutter detection and continuous therapy support. It reduces reliance on manual evaluation, offers objective insights into speech fluency, and can serve as an assistive aid for both speech pathologists and individuals undergoing therapy. This project demonstrates how artificial intelligence and audio signal processing can be effectively utilized in the healthcare domain to enhance diagnostic accuracy, improve therapy outcomes, and promote speech fluency assessment at scale.

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Abbreviation	Description
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GUI	Graphical User Interface
VAD	Voice Activity Detection
STFT	Short-Time Fourier Transform
MFCC	Mel-Frequency Cepstral Coefficients
DADS	Dysfluency Analysis and Detection System
dB	Decibel
Hz	Hertz
PDF	Portable Document Format
IDE	Integrated Development Environment
SLP	Speech-Language Pathologist
HMM	Hidden Markov Model
SVM	Support Vector Machine
ReLU	Rectified Linear Unit
FC	Fully Connected (Layer)
ST	Stutter Type
STT	Speech-to-Text
UI	User Interface
OS	Operating System
PyQt5	Python Qt Framework (for GUI development)
Sep	28k Stuttering Event Prediction Dataset (28,000 samples)
TTS	Text-to-Speech
ML	CNN Multi-Label Convolutional Neural Network
Sigmoid	Activation function for binary classification
Z	Score Statistical normalization technique
n_fft	Number of Fast Fourier Transform points
hop_length	Step size between successive STFT frames
n_mels	Number of Mel filters used for spectrogram generation
VTU	Visvesvaraya Technological University

